DHSK College

Department of Computer Science

BCA – Semester IV

Data Structures

Practical Assignment Booklet

Subject	Data Structures (BCA – Sem IV)
Units Covered	Unit 1 (Arrays, Searching, Sorting) & Unit 2 (Linked Lists, Stacks, Queues)
Programming Language	C
Total Practicals	12 (including 1 Mini Project)
Submission Mode	Google Classroom: (1) Handwritten program – scanned/photographed, (2) Screenshot of output, (3) Viva answers in Word/PDF
Submission Deadline	To be announced on Google Classroom
Assigned by	HOD, Department of Computer Science, DHSK College

Instructions for Students

- Each student must write all programs individually in their practical notebook. Group submissions will not be accepted.
- Write the program neatly by hand in your practical notebook. Clearly write the Practical number, Aim, Code, and Output.
- Scan or clearly photograph each handwritten practical page and upload it as a PDF or image file.
- Run the program on a computer, take a clear screenshot of the output, and upload it along with your handwritten copy.
- Answer all Viva Questions at the end of each practical in a Word or PDF document and upload it as a single combined file named: VivaAnswers_RollNo.pdf

- Naming convention: Practical1_Scan_RollNo.pdf, Practical1_Output_RollNo.png, VivaAnswers_RollNo.pdf
- Plagiarism in handwriting or viva answers will result in cancellation of the entire assignment. Write in your own words.
- Deadline is strictly enforced. Late submissions will not be accepted unless prior permission is obtained.
- For any queries, post a comment on Google Classroom — do not message personally.

Practical List & Marks Distribution

No.	Title	Topic / Unit	Marks
1	Array Operations – Insertion, Deletion & Traversal	Arrays – Unit 1	5
2	Searching – Linear Search & Binary Search	Arrays / Searching – Unit 1	5
3	Sorting – Bubble Sort & Selection Sort	Arrays / Sorting – Unit 1	5
4	Singly Linked List – Create, Insert & Delete	Linked Lists – Unit 2	5
5	Doubly Linked List – Insert & Delete	Linked Lists – Unit 2	5
6	Stack Implementation Using Array	Stacks – Unit 2	5
7	Stack Application – Parenthesis Matching	Stacks / Applications – Unit 2	5
8	Queue Implementation Using Array	Queues – Unit 2	5
9	Circular Queue Implementation	Queues – Unit 2	5

No.	Title	Topic / Unit	Marks
10	Stack Using Linked List	Stacks / Linked Lists – Unit 2	5
11	Queue Using Linked List	Queues / Linked Lists – Unit 2	5
12	Mini Project – Student Record Management Using Linked List	Arrays / Linked Lists (Integration) – Unit 1 & 2	10

Total Marks: 65 (Practicals 1–11: 5 marks each = 55 | Practical 12 Mini Project: 10 marks)

Practical 1: Array Operations – Insertion, Deletion & Traversal

Unit: Unit 1 | Topic: Arrays

Aim	Write a C program to perform insertion, deletion, and traversal operations on a 1D array.

Objectives

- Understand array declaration and initialization in C.
- Implement insertion of an element at a given position.
- Implement deletion of an element from a given position.
- Display the array before and after each operation.

Steps to Follow

1. Declare an integer array of size 10 and take input from the user.
2. Display the original array.
3. Insert an element at a user-specified position.
4. Delete an element from a user-specified position.
5. Display the final array after both operations.

Reference Code (Study & Write in Notebook by Hand)

Note: Do NOT copy-paste. Understand the logic and write the complete program in your own handwriting.

```
#include<stdio.h>
#define MAX 10
void display(int a[], int n) {
    for(int i=0; i<n; i++) printf("%d ", a[i]);
    printf("\n");
}
void insert(int a[], int *n, int pos, int val) {
    for(int i=*n; i>pos; i--) a[i]=a[i-1];
    a[pos]=val; (*n)++;
}
void delete(int a[], int *n, int pos) {
    for(int i=pos; i<*n-1; i++) a[i]=a[i+1];
```

```
    (*n)--;
}
int main() {
    int a[MAX], n;
    printf("Enter n: "); scanf("%d",&n);
    for(int i=0;i<n;i++) scanf("%d",&a[i]);
    display(a,n);
    // Insert and delete as per user input
    return 0;
}
```

Expected Output

Show array before insertion, after insertion, and after deletion.

Viva Questions

Q1. What is the time complexity of insertion at the beginning of an array?

Q2. Why does deletion at position 0 require shifting all elements?

Practical 2: Searching – Linear Search & Binary Search

Unit: Unit 1 | Topic: Arrays / Searching

Aim	Implement and compare Linear Search and Binary Search algorithms on an integer array.

Objectives

- Understand the concept of searching in arrays.
- Implement Linear Search and Binary Search in C.
- Count and compare the number of comparisons made by each algorithm.
- Understand when Binary Search requires sorted data.

Steps to Follow

6. Create an array of 10 integers.
7. Implement Linear Search and display the number of comparisons.
8. Sort the array (use Bubble Sort or built-in sort).
9. Implement Binary Search on the sorted array.
10. Display whether the element was found and how many comparisons were needed.

Reference Code (Study & Write in Notebook by Hand)

Note: Do NOT copy-paste. Understand the logic and write the complete program in your own handwriting.

```
#include<stdio.h>
int linearSearch(int a[], int n, int key) {
    for(int i=0; i<n; i++)
        if(a[i]==key) return i;
    return -1;
}
int binarySearch(int a[], int n, int key) {
    int lo=0, hi=n-1;
    while(lo<=hi) {
        int mid=(lo+hi)/2;
        if(a[mid]==key) return mid;
        else if(a[mid]<key) lo=mid+1;
        else hi=mid-1;
    }
    return -1;
}
```

Expected Output

Position of the element in the array, and number of comparisons for each method.

Viva Questions

Q1. What is the best and worst case time complexity of Binary Search?

Q2. When would you prefer Linear Search over Binary Search?